An Introduction to Deligne's Riemann–Hilbert Correspondence

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Introduction

This note mainly follows P. Deligne's *Equations différentielles à points singuliers réguliers* [Del70] and briefly introduces a generalization of the Riemann–Hilbert correspondence.

In Section 1, we describe the basic ideas and the steps involved in constructing the theory. In Section 2, we give details of the proof of Deligne's Riemann–Hilbert correspondence in the one-dimensional case, except for a detailed discussion of the application of Serre's GAGA theorem. In Section 3, we sketch the proof in the n-dimensional case. Since this is an introductory note, some arguments are intended to convey intuition rather than to serve as complete proofs.

1 General Idea

Suppose that X is a complex manifold. We denote by $Conn(X)$ the category of integrable connections $(\mathscr{V}, \nabla)$ on X. Here $\mathscr{V}$ is a finite-rank holomorphic vector bundle. The connection ∇ satisfies $\nabla^2 = 0$. We denote by $Loc(X)$ the category of local systems (locally constant sheaves) of finite-dimensional complex vector spaces.

The classical Riemann–Hilbert correspondence then asserts that [Del70, Chapter I]

Theorem 1. (Riemann–Hilbert Correspondence)

The categories $Conn(X)$ and $Loc(X)$ are equivalent, and the equivalence is given by the following functors:

(a) Send $V \in Loc(X)$ to the vector bundle $\mathscr{V} = V \otimes_{\mathbb{C}} \mathscr{O}_X$ endowed with its canonical connection ∇, characterized by $\nabla v = 0$ for every local section $v \in V$.

(b) Send $(\mathscr{V}, \nabla)$ to the local system of its horizontal local sections.

We now want to generalize this correspondence to the algebraic setting. Let X be a smooth algebraic variety over $\mathbb{C}$, and denote $\Omega_X^\bullet \otimes \mathscr{V}$ by $\Omega_X^\bullet(\mathscr{V})$.

Definition 1. *Just as in complex geometry, an integrable connection on a locally free coherent sheaf $\mathscr{V}$ over X is a family of $\mathbb{C}$-linear homomorphisms*

$$\nabla \colon \Omega_X^i(\mathscr{V}) \longrightarrow \Omega_X^{i+1}(\mathscr{V})$$

such that, for every local section $\alpha \in \Omega_X^i$ *and* $v \in \mathscr{V}$*,*

(i) the Leibniz rule holds:

$$\nabla(\alpha \otimes v) = d\alpha \otimes v + (-1)^i \alpha \wedge \nabla v;$$

(ii) the connection is integrable, or has zero curvature: $\nabla^2 v = 0$.

We may also view ∇ as a $\mathbb{C}$-linear homomorphism

$$T_X \longrightarrow End(\Omega_X^\bullet(\mathscr{V})), \qquad \tau \longmapsto \nabla_\tau.$$

As in differential geometry, a formal calculation shows that (ii) is equivalent to

(iii) $\nabla_{[\tau_1,\tau_2]} = [\nabla_{\tau_1}, \nabla_{\tau_2}]$

We also use $Conn(X)$ to denote the category of algebraic integrable connections. We might expect an equivalence between $Conn(X)$ and $Conn(X^{an})$, where X^{an} denotes the analytic complex space corresponding to X. Unfortunately, this is false in general. For example, it can fail when X is not projective. In a loose sense, the algebraic category can distinguish "more" connections than the analytic category does. Two non-isomorphic algebraic connections may become isomorphic analytically.

Example 1. *Set* $X = \mathrm{Spec}\mathbb{C}[z, z^{-1}]$*, and consider two algebraic connections on* $\mathscr{O}_X$*:* ∇_1 *is the trivial connection,* $\nabla_1 f = df$*, while*

$$\nabla_2 f = df + f\frac{dz}{z^2}.$$

They are both integrable connections, but they are not isomorphic in the algebraic category. If we pass to the analytic category, however, they become isomorphic.

The connection ∇_1 *has a global nowhere-vanishing horizontal section, namely the* *constant section* $f = 1$*, so it corresponds to the trivial local system of rank one. On the other hand,* ∇_2 *also has a global nowhere-vanishing horizontal section* $f = e^{1/z}$*, which solves*

$$\partial_z f + f/z^2 = 0$$

Therefore, by the analytic Riemann–Hilbert correspondence, ∇_2 *is analytically isomorphic to* ∇_1*.*

Thus we should consider algebraic connections satisfying an additional condition. Deligne gives the appropriate restrictive condition, called *regularity*, which expresses a growth condition at infinity.

The basic idea behind this condition is the following.

On a projective variety, GAGA allows us to compare analytic and algebraic objects [Ser56]. We therefore try to extend connections on X meromorphically to a compactification $\bar{X}$. Consider algebraic connections on $\bar{X}$ that are meromorphic along $\bar{X} \setminus X$. Their category is naturally related to $Conn(X)$. We now need an additional condition. It should identify the meromorphic analytic connections on $\bar{X}$ that correspond canonically to connections on X^{an}.

Take a meromorphic connection on $\bar{X}$ with poles along $\bar{X} \setminus X$. Its restriction to X is a holomorphic analytic connection. Hence the desired condition must distinguish which meromorphic extensions behave in the most "algebraic" way. In other words, it should identify the extensions with the kind of behavior one expects from algebraic geometry.

This gives an intuitive picture of the condition we are looking for. After choosing a suitable extension, local horizontal sections should behave "algebraically." In particular, their growth near the boundary should resemble the growth of meromorphic functions. It should not resemble the growth of a function such as $e^{1/z}$. This observation will eventually be formulated as the regularity condition.

Ideas from the theory of ODEs lead to a more computable condition. The regularity condition turns out to be equivalent to this new condition. Namely, in a suitable extension, the connection forms have at worst logarithmic poles at infinity [Del70, Chapter II]. Roughly speaking, such poles resemble df/f.

Now suppose that $\bar{X} \setminus X$ is a simple normal-crossings (SNC) divisor. Hironaka's resolution theorem allows us to reach this situation [Hir64a, Hir64b]. One can then show that every connection in $Conn(X^{an})$ admits a canonical logarithmic extension [Del70, Chapter II, Section 5]. It is unique after an appropriate choice of residue representatives. Without that choice, it is unique only up to the relevant meromorphic equivalence.

Although this description is still somewhat vague, we can use the preceding example to illustrate the phenomenon.

Example 2. *As in Example 1, set* $X = \mathrm{Spec}\mathbb{C}[z, z^{-1}]$ *and let* $\bar{X} = \mathrm{Spec}\mathbb{C}[z]$*, a partial compactification of* X *obtained by adding the point* 0*. It is no surprise that the trivial connection on* $\bar{X}$ *gives an extension of* ∇_1*. But we can also calculate the connection form of* ∇_1 *using the frame (a nowhere-vanishing section on* X*)* $s_k = z^k$*. Each such choice corresponds to a different lattice extending* $\mathscr{O}_X$ *across* 0*, and*

$$\nabla_1 s_k = k\frac{dz}{z} \otimes s_k.$$

Thus these extensions have only logarithmic poles and should also be regarded as regular.

For ∇_2, calculating the connection form in the frame $s_k = z^k$ gives

$$\nabla_2 s_k = (kz^{-1} + z^{-2})dz \otimes s_k$$

The connection form $(kz^{-1} + z^{-2})dz$ therefore always contains a second-order pole at 0*; multiplying the frame by a power of z cannot remove the $z^{-2}dz$ term. Thus ∇_2 is ruled out by the regularity condition.*

In the following sections, we will try to formulate the ideas above precisely. First, keeping the rough intuition obtained from these examples in mind, let us concentrate on the one-dimensional case.

2 Regularity in Dimension 1

The first step is to define meromorphic connections and regularity.

We begin with the analytic case. Let X be the punctured unit disk and $\bar{X}$ the open unit disk, and write $j\colon X \hookrightarrow \bar{X}$ for the inclusion. Viewing 0 as a point at infinity of X, we study extensions of a connection $(\mathscr{V}, \nabla) \in Conn(X)$ near 0.

The calculation in Example 2 suggests changing a rank-one frame by multiplying it by z^k. Under this change, the connection form merely acquires an additional term $k\, dz/z$. The change does not affect the existence of a frame in which the connection has at worst a logarithmic pole. It is therefore natural to introduce the following equivalence relation on extensions.

Definition 2. *Two extensions (or lattices) $\widetilde{\mathscr{V}_1}$ and $\widetilde{\mathscr{V}_2}$ of $\mathscr{V}$ across* 0 *are equivalent* *near* 0 *if there exists an integer* $k \geq 0$ *such that*

$$z^k \widetilde{\mathscr{V}}_1 \subseteq \widetilde{\mathscr{V}}_2 \subseteq z^{-k} \widetilde{\mathscr{V}}_1 \subseteq j_* \mathscr{V}$$

We say that a vector bundle on X*, equipped with such an equivalence class of extensions near* 0*, is a meromorphic vector bundle near* 0*.*

Choose a fixed extension in the equivalence class of a rank-n meromorphic vector bundle $\mathscr{V}$ near 0. Locally, a frame $s_1, \ldots, s_n$ on X represents this extension. After trivializing the extension, we identify its lattice with $\mathscr{O}_{\bar{X}}^n$. Under this identification, s_i is sent to the constant section of the i-th summand $\mathscr{O}_{\bar{X}} \subseteq \mathscr{O}_{\bar{X}}^n$.

In this frame, we can calculate the connection matrix $\Gamma = (\Gamma_i^j)$ of ∇. It is defined by $\nabla s_i = \sum_j \Gamma_i^j \otimes s_j$. Suppose the entries of Γ are meromorphic differential forms on $\bar{X}$. We then say that the meromorphic vector bundle equipped with ∇ is a meromorphic connection near 0. Under a gauge transformation A, the connection matrix changes according to

$$\Gamma' = A^{-1} \Gamma A + A^{-1} dA$$

Since a change between equivalent lattices is meromorphic, this condition is independent of the chosen lattice and frame. Notice, however, that the *order* of the pole of a displayed connection matrix need not be preserved by an arbitrary meromorphic gauge transformation.

We can now define regularity.

Definition 3. *A meromorphic connection* $(\mathscr{V}, \widetilde{\mathscr{V}}, \nabla)$ *is regular singular near* 0 *if a suitable lattice has the following property. There is a basis of the lattice in which the* *entries Γ_i^j have at worst simple poles at* 0*. Equivalently, for some lattice $\widetilde{\mathscr{V}}$,*

$$\nabla(\widetilde{\mathscr{V}}) \subseteq \Omega^1_{\bar{X}}(\log 0) \otimes \widetilde{\mathscr{V}}.$$

We now turn to the algebraic case. Let X be a smooth affine algebraic curve over $\mathbb{C}$. Let $\bar{X}$ be the smooth projective curve containing it. Choose a point $0 \in \bar{X} \setminus X$ and a uniformizer z near 0. We regard 0 as a point at infinity of X. We will study extensions of an algebraic connection $(\mathscr{V}, \nabla) \in Conn(X)$ near 0.

The preceding definition of a meromorphic vector bundle can be imitated directly. In the algebraic one-dimensional case, any two lattices in the same generic fiber are commensurable, so all extensions of a given vector bundle determine the same meromorphic equivalence class.

Next, let us explain how the connection matrix is defined. Denote by K the fraction field of the local ring $\mathscr{O} = \mathscr{O}_{0,\bar{X}}$. Consider a meromorphic vector bundle of rank n near 0. Suppose that $\mathscr{V}$ can be trivialized over a neighborhood $0 \in U = \operatorname{Spec} A$ by sections $s_1, \ldots, s_n \in A_z$. A lattice represents a choice of extension. For example, we can use the inclusion $A^n \hookrightarrow A_z^n$. Passing to the local ring gives an inclusion $\mathscr{O}^n \hookrightarrow K^n$. This inclusion represents the lattice near 0. Let Ω^1_K denote the module of differentials of K. Then ∇ extends naturally to an additive map, still denoted by ∇,

$$\nabla \colon K^n \longrightarrow \Omega^1_K \otimes_K K^n.$$

This construction enables us to calculate the connection matrix Γ of ∇ relative to a meromorphic extension. We use a basis $s_1, \ldots, s_n$ of K^n. The entries of Γ are automatically meromorphic. We can therefore define regularity exactly as in the analytic case. The connection is regular at 0 if some lattice gives a matrix with at worst a simple pole.

From the discussion above, regularity is an intrinsic property of an algebraic connection, at least in the one-dimensional case. Hence there is a full subcategory of $Conn(X)$ consisting of regular algebraic connections; we denote it by $Conn^{reg}(X)$.

We now complete the one-dimensional program proposed in Section 1. We must verify that every analytic connection has a regular meromorphic structure. We must also show that this structure is unique up to meromorphic equivalence. If we choose a fundamental strip for the residue eigenvalues, we obtain a distinguished lattice rather than only an equivalence class.

The extension problem is local, at least in dimension one. It is therefore enough to take X to be the punctured unit disk and $\bar{X}$ the open unit disk. We then construct a regular extension of an analytic connection $(\mathscr{V}, \nabla)$ from X to $\bar{X}$.

We are going to use the Riemann–Hilbert correspondence to construct the extension.

Let $(\mathscr{V}, \nabla)$ be a connection on X. Every element of the fundamental group $\pi_1(X) \cong \mathbb{Z}$ acts on a fiber of $\mathscr{V}$. The action is given by parallel transport along a representing loop. Choose a positively oriented generator of $\pi_1(X)$ and denote the associated action by T. This is the monodromy transformation of ∇. Changing the base point or the basis of the fiber changes the matrix of T by conjugation. Thus its conjugacy class is intrinsic. Let us consider some examples.

Example 3. *Suppose* $\mathscr{V} = \mathscr{O}_X^n$ *and the connection matrix of* ∇ *is* $U\,dz/z$ *for* $U \in$

$M_n(\mathbb{C})$. A local horizontal section s satisfies $\partial_z s = -U \cdot s/z$, so it has the form

$$s = \exp(-(\ln z)U) \cdot v$$

where v is a fixed vector. When z travels once counterclockwise around 0, $\log z$ changes to $\log z + 2\pi i$. Thus the monodromy is the constant matrix

$$T = \exp(-2\pi i U),$$

not the matrix U itself.

Example 4. *Suppose $\mathscr{V} = \mathscr{O}_X^2$ and, under the convention $\partial_z v = -\Gamma v$, the connection matrix is*

$$\begin{pmatrix} 0 & 0 \\ -1 & -1/z \end{pmatrix}$$

A horizontal section has the form

$$\begin{pmatrix} a \\ az\ln z + bz \end{pmatrix}$$

for constants a, b. Analytic continuation around 0 sends (a, b) to $(a, b + 2\pi i a)$, so, in the parameter basis (a, b), the monodromy transformation is

$$\begin{pmatrix} 1 & 0 \\ 2\pi i & 1 \end{pmatrix}$$

The examples suggest a relation between the monodromy T and the residue of the connection matrix Γ. The precise statement requires some care. The second example is resonant: its residue eigenvalues differ by a nonzero integer. In this situation, holomorphic terms can contribute to the unipotent part of monodromy. A statement about the eigenvalues is always available. For a nonresonant logarithmic lattice, it strengthens to a statement about the full conjugacy class. This applies in particular to Deligne's canonical lattice introduced below.

Theorem 2. *Let* $R = \mathrm{Res}_0(\Gamma)$ *be the residue of a logarithmic connection. The eigenvalues of its local monodromy are* $\exp(-2\pi i\rho)$*, where* ρ *runs through the eigenvalues of* R*, with multiplicity. If no two eigenvalues of* R *differ by a nonzero integer, then the monodromy is conjugate to* $\exp(-2\pi i R)$ *[Del70, Chapter II].*

Proof. The following calculation explains the intuition. Write the connection form as

$$\Gamma = A(z)\,dz, \qquad A(z) = \frac{R}{z} + B(z),$$

where $R = \mathrm{Res}_0(\Gamma)$ and $B(z)$ is holomorphic. A horizontal section satisfies

$$\partial_z v = -A(z)v.$$

Along the circle $z = re^{i\theta}$, the chain rule gives

$$\partial_\theta v = -izA(z)v = -iRv - izB(z)v.$$

The last term tends uniformly to zero as $r \to 0$ with θ in a compact interval. The limiting equation is therefore

$$\partial_\theta v = -i\mathrm{Res}(\Gamma) \cdot v$$

This calculation explains why the exponential of the residue controls the monodromy eigenvalues. It is still only a heuristic for the full conjugacy statement. A rigorous proof uses the standard theory of regular singular differential equations [Del70, Chapter II]. In the resonant case, holomorphic terms can contribute to the unipotent part of monodromy. In the nonresonant case, the result reduces to the model calculation in Example 3. $\square$

Let us return to the construction of a regular extension. For any connection $(\mathscr{V}, \nabla)$, choose $U \in M_n(\mathbb{C})$ such that $\exp(-2\pi i U)$ is conjugate to the monodromy T. The connection $(\mathscr{O}_X^n, \nabla')$ constructed in Example 3 is regular with respect to the trivial extension. The analytic Riemann–Hilbert correspondence tells us that two flat connections on the punctured disk with the same monodromy representation are isomorphic. We transport the trivial logarithmic lattice through such an isomorphism. This gives a regular meromorphic extension of ∇.

We may want a particular lattice rather than merely a meromorphic equivalence class. In that case, choose a set $\tau \subset \mathbb{C}$ containing one representative of each class in $\mathbb{C}/\mathbb{Z}$. For example, take $\tau = \{a \in \mathbb{C} : 0 \leq \mathrm{Re}(a) < 1\}$. Requiring the eigenvalues of U to lie in τ singles out Deligne's canonical extension [Del70, Chapter II, Section 5].

We now explain the relation between the regularity condition defined above and the growth condition that motivated it. The uniqueness of the regular meromorphic structure is also contained in this discussion.

Definition 4. *Let f be holomorphic on a sector U approaching 0. We say that f has* moderate growth *if there are constants $A > 0$ and $n \geq 0$ such that*

$$|f(z)| \leqslant A|z|^{-n}$$

for z sufficiently close to 0*. A local, possibly multivalued, section of a meromorphic bundle has moderate growth if, in some meromorphic frame and on every sufficiently small closed sector on which a branch is chosen, all of its coordinate functions have moderate growth at* 0*.*

The use of sectors is necessary because horizontal sections may involve $\log z$ and hence may be multivalued. The condition does not depend on the meromorphic frame, since a meromorphic change of frame changes growth by at most a power of $|z|$.

Theorem 3. *Let* $(\mathscr{V}, \nabla)$ *be a meromorphic connection near* 0 *on the punctured unit disk, then the following conditions are equivalent:*

(i) $(\mathscr{V}, \nabla)$ *is regular.*

(ii) the local horizontal sections of $(\mathscr{V}, \nabla)$ *are of moderate growth near* 0 *on sectors.*

[Del70, Chapter II, Section 1]

Proof. (i) $\Rightarrow$ (ii). Since ∇ is regular, choose a frame in which

$$\Gamma = \left(\frac{R}{z} + B(z)\right) dz,$$

with $B(z)$ holomorphic. The standard local normal-form theorem for regular singular systems shows that, on a sector, the coordinates of a horizontal section are finite sums of terms of the form

$$z^{\lambda}(\log z)^{q} h(z),$$

where $q \geq 0$ and h is holomorphic near 0 [Del70, Chapter II, Section 1]. On every closed sector, $|z^{\lambda}|$ is bounded by a fixed power of $|z|^{-1}$. Powers of $\log z$ are also bounded by a slightly larger power of $|z|^{-1}$. Hence every horizontal section has moderate growth.

(ii) $\Rightarrow$ (i). Suppose that, in a meromorphic frame $s_1, \ldots, s_n$, the local horizontal sections of $(\mathscr{V}, \nabla)$ have moderate growth. Apply the preceding construction, based on the Riemann–Hilbert correspondence, to obtain an isomorphism

$$\varphi\colon (\mathscr{V}, \nabla) \longrightarrow (\mathscr{O}_X^n, \nabla')$$

of flat connections on X, where ∇' is logarithmic. It remains to show that this isomorphism is meromorphic at 0, so that it respects the meromorphic structures.

Let A be the matrix of φ from the frame $s_1, \ldots, s_n$ to the constant frame $e_1, \ldots, e_n$ of $\mathscr{O}_X^n$. Let M be the fundamental matrix in the frame s_i. Its columns are the coordinates of a local horizontal basis $v_1, \ldots, v_n$. Let M' be the corresponding matrix in the frame e_i. Its columns are the coordinates of $\varphi(v_1), \ldots, \varphi(v_n)$. Up to the convention used for coordinate vectors, we have locally $A = M'M^{-1}$.

The idea is that the entries of both fundamental matrices, together with the entries needed for their inverses, have at most polynomial growth. Then A and A^{-1} have at most polynomial growth, and a single-valued holomorphic function with such growth extends meromorphically across 0. Thus φ identifies the given meromorphic structure with a logarithmic one.

There is a small technical point hidden here: moderate growth of the entries of M alone does not automatically imply moderate growth of M^{-1} by elementary linear algebra. A complete proof uses the dual connection, or equivalently the standard regular-singular criterion for a system and its dual. Since the purpose of this note is to explain the idea, we leave this standard analytic step implicit. $\square$

Moreover, the proof shows that every regular extension of a connection is equivalent to the one constructed above, which gives uniqueness of the regular meromorphic structure. We can also see that the construction is functorial and fully faithful by the preceding theorem.

Indeed, consider a morphism φ between two regular connections $(\mathscr{V}, \widetilde{\mathscr{V}}, \nabla)$ and $(\mathscr{V}', \widetilde{\mathscr{V}}', \nabla')$. We may view it as a global horizontal section of $\mathrm{Hom}(\mathscr{V}, \mathscr{V}')$. This bundle carries the connection induced by ∇ and ∇'. The induced connection is again regular. Its natural lattice is obtained from $\mathrm{Hom}(\widetilde{\mathscr{V}}, \widetilde{\mathscr{V}}')$. By the same moderate-growth argument, φ extends meromorphically. After passing to a projective compactification, this is the type of statement that can be algebraized by GAGA.

Eventually, let X be a smooth affine algebraic curve. We obtain a functor from $Conn(X^{an})$ to $Conn^{reg}(X)$. First take the regular extension at every point at infinity. Then apply GAGA.

Strictly speaking, the applicability of GAGA must be formulated carefully. A connection is $\mathbb{C}$-linear rather than $\mathscr{O}_X$-linear. One can package the connection together with coherent sheaves and compatible morphisms. Alternatively, one can work in the corresponding $\mathscr{D}$-module framework. We refer the reader to Chapter 5 of *D-Modules, Perverse Sheaves, and Representation Theory* by R. Hotta, K. Takeuchi, and T. Tanisaki for details [HTT08, Chapter 5]. The discussion above suggests that the resulting functor is fully faithful and essentially surjective. Hence it is an equivalence.

In the n-dimensional case, the steps used to construct the correspondence are very similar to those in the one-dimensional case, but several points become much more subtle. We sketch them in the next section.

3 The Proof in Higher Dimensions

We try to follow the same process as in the one-dimensional case.

In order to use GAGA, we make one of two assumptions. Either X is the analytification of a smooth quasi-projective algebraic variety, or we work directly with such an algebraic variety. Hironaka's resolution theorem gives a smooth projective compactification $\bar{X}$ of X [Hir64a, Hir64b]. We can choose it so that $Y = \bar{X} \setminus X$ is a simple normal-crossings divisor. We assume throughout this section that $\bar{X}$ is of this form.

The definition of a meromorphic connection from Section 2 still works in the higher-dimensional algebraic setting. We denote by $Conn(\bar{X}, Y)$ the category of locally free $\mathscr{O}_{\bar{X}}(*Y)$-modules of finite rank equipped with integrable connections. Informally, these are connections on $\bar{X}$ that are meromorphic along Y. When X is algebraic, restriction to X relates $Conn(\bar{X}, Y)$ to $Conn(X)$. A precise equivalence uses the standard extension theorem for coherent $\mathscr{O}_{\bar{X}}(*Y)$-modules. We suppress this technical point in this note.

We next define regularity in the algebraic setting. Let X be a smooth algebraic variety and C a smooth algebraic curve. Given a connection $(\mathscr{V}, \nabla)$ on X, a morphism $i\colon C \to X$ induces a connection $i^*\nabla$ on $i^*\mathscr{V}$.

Definition 5. *We say that $(\mathscr{V}, \nabla)$ is regular if, for every smooth algebraic curve C and every morphism $i\colon C \to X$, the induced connection $(i^*\mathscr{V}, i^*\nabla)$ is regular on C in the sense of Section 2.*

This is an intrinsic property of a connection. We denote by $Conn^{reg}(X)$ the full subcategory of $Conn(X)$ consisting of all regular algebraic connections on X, and by $Conn^{reg}(\bar{X}, Y)$ the corresponding category of regular meromorphic connections on the compactification.

We turn to the analytic case now.

The first subtle problem in the n-dimensional analytic case concerns meromorphic structures on analytic connections. By analogy with the one-dimensional case, we can define such a structure locally as an equivalence class of extensions. There is no immediate reason for these local classes to glue to a global extension on $\bar{X}$. To match the algebraic setting, we restrict attention to structures represented by a global coherent lattice. These are called *effective meromorphic connections*. Their category is denoted by $Conn^e(\bar{X}, Y)$.

The regularity condition in the analytic setting can be defined similarly to the algebraic one.

Definition 6. *Fix a compactification $\bar{X}$. Consider a meromorphic connection $(\mathscr{V}, \widetilde{\mathscr{V}}, \nabla) \in Conn(\bar{X}, Y)$. We say that it is regular if it passes the following test. Let i be any nonconstant holomorphic map from the unit disk $D = \{z \in \mathbb{C} : |z| < 1\}$ to $\bar{X}$. Assume that $i(D^*) \subset X$ and $i(0) \in Y$. Then the induced meromorphic connection on D must be regular at 0. We denote the resulting category by $Conn^{reg}(\bar{X}, Y)$.*

This disk test is the analytic analogue of the algebraic curve test. Their agreement is not a formal consequence of the definitions. It is part of Deligne's regularity criterion and uses compactification and resolution arguments; see [Del70, Chapter II] and [And07].

At first sight, neither $Conn^e(\bar{X}, Y)$ nor $Conn^{reg}(\bar{X}, Y)$ appears independent of the chosen compactification. Nevertheless, an appropriate GAGA comparison relates $Conn^e(\bar{X}^{an}, Y^{an})$ to $Conn(\bar{X}, Y)$ [Ser56, HTT08], and regularity is preserved under this comparison. The eventual correspondence on the open variety X will therefore be intrinsic.

As in Section 2, one of the most important steps is to construct the canonical extension of an analytic connection. There is an important distinction to keep in mind. We directly construct an extension with logarithmic poles. We do not directly construct an extension satisfying only the curve-test definition of regularity. After the construction, we will explain why these two notions are equivalent. The logarithmic-pole condition is a concrete condition on a chosen lattice. By contrast, regularity was defined intrinsically by testing curves.

Suppose that, after imposing a suitable condition, we can prove uniqueness as a specific extension. This is stronger than uniqueness of an equivalence class. The gluing problem then becomes much easier. Unique local extensions must agree on overlaps. It is therefore enough to solve the problem locally. Near a point where exactly r components of Y meet, take $\bar{X}$ to be the unit polydisk

$$\{(z_1, \ldots, z_m) \in \mathbb{C}^m : |z_i| < 1 \text{ for all } i\},$$

with $Y = \{z_1 \cdots z_r = 0\}$. Then X is locally $(D^*)^r \times D^{m-r}$. Denote by Y_i the divisor cut out by z_i for $1 \leq i \leq r$. The original special case $r = m$ contains the essential monodromy calculation, while this slightly more general notation describes an arbitrary SNC boundary point.

Just as in the one-dimensional case, we can define monodromy around each Y_i. For a connection of rank n on X, call this transformation T_i. Locally, $\pi_1(X) \cong \mathbb{Z}^r$. Hence the commuting transformations $T_1, \ldots, T_r$ determine the isomorphism class of the local system. They also determine the flat analytic connection. Guided by

Section 2, we choose matrices $U_i \in M_n(\mathbb{C})$ such that

$$\exp(-2\pi i U_i) = T_i.$$

There are many possible logarithms. They differ essentially by choices of integers on generalized eigenspaces. Because we want uniqueness, we must specify a rule for choosing them. For example, we may require every eigenvalue of every U_i to have real part in $[0, 1)$.

There is one more condition that is easy to overlook. Since the T_i commute, we choose the U_i by applying compatible branches of the logarithm to the T_i. The resulting U_i also commute. This is essential: without $[U_i, U_j] = 0$, the model connection below would not be integrable.

We then construct a connection ∇' on $\mathscr{O}_X^n$ by taking its connection matrix to be

$$\Gamma = \sum_{i=1}^{r} U_i \frac{dz_i}{z_i}.$$

The solution of this connection is of the form

$$\exp(-\sum (\ln z_i) U_i) \cdot v$$

where v is a constant vector. Because the U_i commute, this expression is unambiguous. It shows that ∇' has the same monodromy as ∇. The model has a trivial lattice. Transporting this lattice through the analytic Riemann–Hilbert isomorphism gives the desired logarithmic extension.

The various choices of U_i prevent uniqueness. We therefore impose an additional condition on the residue matrices along Y_i, denoted by $\mathrm{Res}_i(\nabla)$. Their eigenvalues must have real parts in $[0, 1)$. More generally, one may choose any set containing exactly one representative of each class in $\mathbb{C}/\mathbb{Z}$.

Suppose now that there are two logarithmic extensions of ∇, both satisfying this residue condition. Let their connection matrices be Γ and Γ'. The meromorphic gauge transformation A between them satisfies

$$A\Gamma' = \Gamma A + dA$$

We want to prove that A extends holomorphically and invertibly across Y. First work near a general point of one component Y_k. Stay away from the intersections with the other components. The problem then reduces to the one-dimensional transverse direction. The remaining subset has codimension at least two and is handled by an extension theorem.

The one-dimensional discussion shows that A is meromorphic. Suppose that A has a pole of order $k > 0$. We examine its first nonzero Laurent coefficient A_{-k}. Compare the lowest-order terms in the gauge equation. They show that A_{-k} induces a map between generalized residue eigenspaces. The corresponding eigenvalues differ by the positive integer k. This contradicts the assumption that all residue eigenvalues lie in the same fundamental strip. Apply the same argument to A^{-1}. It follows that A is holomorphic and invertible.

It is tempting to write

$$\mathrm{Res}(\Gamma') = A_{-k}^{-1}(\mathrm{Res}(\Gamma) + kI)A_{-k},$$

as in the original heuristic argument. However, the leading coefficient A_{-k} need not be invertible. The generalized-eigenspace formulation above is the correct version of that intuition.

Definition 7. *Let Y be a simple normal-crossings divisor. A connection on a lattice $\widetilde{\mathscr{V}}$ has* logarithmic poles along Y *if*

$$\nabla(\widetilde{\mathscr{V}}) \subseteq \Omega^1_{\bar{X}}(\log Y) \otimes \widetilde{\mathscr{V}}.$$

Thus, in coordinates for which $Y = \{z_1 \cdots z_r = 0\}$, its connection matrix is a sum of a holomorphic matrix-valued one-form and terms of the form $R_i\, dz_i/z_i$.

We can now state the theorem, due to Deligne, that identifies regularity with the logarithmic-pole condition.

Theorem 4. *For a connection $(\mathscr{V}, \widetilde{\mathscr{V}}, \nabla) \in Conn^e(\bar{X}, Y)$, the following conditions are equivalent:*

(i) ∇ is regular.

(ii) locally, the meromorphic equivalence class contains a lattice in which the connection matrix Γ has at worst logarithmic poles;

(iii) the extension $\widetilde{\mathscr{V}}$ is meromorphically equivalent to the canonical extension $\bar{\mathscr{V}}$ constructed above.

[Del70, Chapter II]

Proof. The implications (iii) $\Rightarrow$ (ii) and (ii) $\Rightarrow$ (i) are clear from the construction and the one-dimensional criterion.

For (i) $\Rightarrow$ (iii), take a general point y of a component Y_k. Choose a tubular neighborhood U of y along Y_k. There is a biholomorphic coordinate map from U to

D^m. It maps Y_k to $\{0\} \times D^{m-1}$. The curve-test assumption implies that the relevant gauge transformations are meromorphic on every transverse disk $D \times \{u\}$. We must show that this fiberwise meromorphic behavior is jointly meromorphic on U.

To see the idea, let f be holomorphic on $U \setminus Y_k$ and meromorphic on every disk $D \times \{u\}$. Write

$$f(z,u) = \sum_{j \in \mathbb{Z}} a_j(u) z^j.$$

The Laurent coefficients $a_j(u)$ are holomorphic in the parameter u. For every fixed u, only finitely many negative coefficients are nonzero. Hence the closed sets

$$F_N = \{u : a_{-j}(u) = 0 \text{ for every } j > N\}$$

cover the parameter space. By Baire's theorem, some F_N has nonempty interior. The identity theorem then implies that a_{-j} vanishes identically for all $j > N$, giving a locally uniform bound on the pole order. Therefore f is meromorphic on U.

Apply this argument to the entries of the gauge transformation. Combine it with the uniqueness of $\bar{\mathscr{V}}$. This gives the desired extension near the general points of Y. A Hartogs-type theorem extends across the intersections of boundary components. Uniqueness on overlaps then completes the gluing argument. We leave this last technical step at the level of a sketch.

□

So far, we have explained the existence and uniqueness of a logarithmic extension after imposing a condition on the residue eigenvalues. Equivalently, we have chosen logarithms of the local monodromy. If we change the representatives in $\mathbb{C}/\mathbb{Z}$, we obtain a different canonical lattice. The theorem shows that all such lattices lie in the same meromorphic equivalence class.

Moreover, regular meromorphic connections admit effective logarithmic representatives. GAGA therefore gives a natural equivalence between $Conn^{reg}(\bar{X}, Y)$ and $Conn^{reg}(\bar{X}^{an}, Y^{an})$ [Del70, Chapter II]. As in Section 2, a complete proof requires a precise GAGA statement for connections or $\mathscr{D}$-modules. Ordinary GAGA for an isolated coherent sheaf is not by itself enough.

Finally, consider the following commutative diagram:

$$\begin{array}{ccc} Conn^{reg}(\bar{X}, Y) & \longrightarrow & Conn^{reg}(\bar{X}^{an}, Y^{an}) \\ \downarrow & & \downarrow \\ Conn^{reg}(X) & \longrightarrow & Conn(X^{an}) \end{array}$$

Combining the horizontal comparison functors with restriction to the open variety gives the correspondence we want.

Theorem 5. (Deligne′s Riemann–Hilbert Correspondence)

Let X be a smooth quasi-projective algebraic variety over $\mathbb{C}$. Then analytification gives an equivalence between $Conn^{reg}(X)$ and $Conn(X^{an})$. Equivalently, after applying the classical analytic Riemann–Hilbert correspondence, there is an equivalence [Del70, Chapter II]

$$Conn^{reg}(X) \simeq Loc(X^{an}).$$

References

[And07] Yves André. An algebraic proof of deligne's regularity criterion for integrable connections, 2007.

[Del70] Pierre Deligne. *Équations différentielles à points singuliers réguliers*, volume 163 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin, Heidelberg, New York, 1970.

[Hir64a] Heisuke Hironaka. Resolution of singularities of an algebraic variety over a field of characteristic zero. i. *Annals of Mathematics*, 79(1):109–203, 1964.

[Hir64b] Heisuke Hironaka. Resolution of singularities of an algebraic variety over a field of characteristic zero. ii. *Annals of Mathematics*, 79(2):205–326, 1964.

[HTT08] Ryoshi Hotta, Kiyoshi Takeuchi, and Toshiyuki Tanisaki. *D-Modules, Perverse Sheaves, and Representation Theory*, volume 236 of *Progress in Mathematics*. Birkhäuser, Boston, 2008.

[Ser56] Jean-Pierre Serre. Géométrie algébrique et géométrie analytique. *Annales de l'Institut Fourier*, 6:1–42, 1956.